

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$Z[J^\mu, \bar{\eta}, \eta] = \int [dA_\mu] \mathfrak{S}(A_\mu) \exp \left(- \left\langle \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right\rangle + \langle J^\mu A_\mu \rangle + \left\langle \bar{\eta}' (i\gamma^\mu \partial_\mu)^{-1} \eta' \right\rangle \right),$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{4!} \int_{\Lambda} \frac{d^4 \vec{p}_1}{(2\pi)^4} \frac{d^4 \vec{p}_2}{(2\pi)^4} \frac{d^4 \vec{p}_3}{(2\pi)^4} \frac{d^4 \vec{p}_5}{(2\pi)^4} \delta \lambda_4(1235) \delta^4(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \vec{p}_5) \hat{\phi}_2(\vec{p}_1) \hat{\phi}_2(\vec{p}_2) \hat{\phi}_2(\vec{p}_3) \hat{\phi}_2(\vec{p}_5).$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial T^{ab}}{\partial w_{ij}} = \frac{\partial T^{ab}}{\partial w_{ji}}, \frac{\partial T^{ab}}{\partial w_{ijk}} = \frac{\partial T^{ab}}{\partial w_{jik}}, \frac{\partial T^{ab}}{\partial w_{ijkl}} = \frac{\partial T^{ab}}{\partial w_{jikl}} = \frac{\partial T^{ab}}{\partial w_{ijlk}} = \frac{\partial T^{ab}}{\partial w_{jilk}}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\int_0^{2\pi} d\sigma X^i(\sigma) \hat{P}_i(\sigma) = x_0^i \hat{p}_0^i + \frac{g_{ij}}{2\sqrt{\alpha'}} \sum_{n=1}^{\infty} \left\{ \left(\chi_n^i \alpha_{-n}^j + \bar{\chi}_n^i \tilde{\alpha}_{-n}^j \right) + \left(\bar{\chi}_n^i \alpha_n^j + \chi_n^i \tilde{\alpha}_n^j \right) \right\}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\bar{\Xi}_{\lambda_2}^{\lambda_1} \otimes l_j^i \otimes 1 \left(\bar{\phi}^{\rho_1} \otimes s^{\dot{K}} \otimes \phi^{\rho_2} \right) = \delta_{\lambda_2}^{\rho_1} \sum_{\dot{K}_1 \dot{K}_2 = \dot{K}} \delta_j^{\dot{K}_1} \bar{\phi}^{\lambda_1} \otimes s^{\dot{K}_2} \otimes \phi^{\rho_2}.$$